MODERN SOCIETAL CHALLENGES ABSTRACTS

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1. English Abstract: Addressing Youth Unemployment, Rural-Urban Disparities, and Cybersecurity Threats

Contemporary Kenyan society stands at a critical crossroads defined by three converging challenges: high youth unemployment among university graduates, severe infrastructural disparities between rural agricultural counties like Kitui and urban centers, and escalating cybersecurity vulnerabilities across digital platforms. This analytical presentation investigates how targeted Information Technology interventions can mitigate these crises. First, by fostering student participation in practical hackathons, machine learning communities (DSAIC), and entrepreneurship, universities can transition graduates from job seekers into tech founders and IT consultants. Second, deploying lightweight, accessible web applications and optimized logistical networks bridges the rural-urban service gap. Finally, embedding Cisco-standard cybersecurity hygiene and secure software architecture into public and private systems is essential to safeguarding national digital infrastructure against data breaches and identity theft.

2. Kiswahili Abstract: Kukabiliana na Ukosefu wa Ajira kwa Vijana, Tofauti za Vijijini na Mijini, na Vitisho vya Mtandaoni

Jamii ya sasa ya Kenya inakabiliwa na changamoto tatu kuu zinazoingiliana: ukosefu mkubwa wa ajira kwa vijana wanaohitimu vyuo vikuu, tofauti kubwa za kimiundombinu kati ya kaunti za vijijini kama Kitui na miji mikuu, na kuongezeka kwa vitisho vya usalama wa mtandao (cybersecurity). Wasilisho hili linachunguza jinsi suluhisho za Teknolojia ya Habari zinavyoweza kutatua changamoto hizi. Kwanza, kwa kuwahusisha wanafunzi katika mashindano ya uundaji programu (hackathons), vilabu vya sayansi ya data (DSAIC), na ujasiriamali, vyuo vikuu vinaweza kuwabadilisha wahitimu kuwa waanzilishi wa kampuni za teknolojia na washauri wa IT. Pili, kuunda mifumo rahisi ya mtandao kunapunguza pengo la huduma kati ya vijiji na miji. Mwisho, kuimarisha usalama wa mifumo ya kompyuta ni muhimu ili kulinda taarifa za wananchi na taasisi dhidi ya udukuzi.

3. Indigenous (Kikamba) Abstract: Kũmina Thĩna wa Wĩra kwa Amũika, Tofauti sya Ndũanĩ na Mĩsyĩnĩ, na Mũisyo wa Mitandao

Nthĩ ya Kenya mĩthenya ĩĩ yĩthĩawa na mathĩna matatũ manene: kũlea kũkwata wĩra kwa amũika ala mamina sukulu nene (universities), tofauti nene ya maendeeo katĩ wa ndũa sya ũĩmi ta Kitui na mĩsyĩ mĩnene, na kwongeleka kwa mũisyo wa ũũi wa mitandaonĩ (cybersecurity threats). Ũvoo ũũ wĩonany'a ũndũ Tekinolojia ya Ũvoo (IT) ĩtonya kũtũmĩka kũmina mathĩna aa. Mbee, kwa kũthuthya amanyĩw'a malike nthĩnĩ wa hackathons, ikundi sya Data Science (DSAIC), na biashara, amũika nĩmatonya kwambĩĩsya kambuni syoo sya tekinolojia vandũ va kwĩkala maendeee kũmantha wĩra. Kelĩ, kũseũvya programu sya mtandao nĩkũtethasya kũete ũtethyo ndũanĩ. Mũthya, kũtũmĩa ũmanyi wa Cisco Cybersecurity kũsũvĩa mĩvango ya kompiuta nĩ kwa vata mũno nĩ kenda ndũmĩrĩri na mali sya andũ iikavũthĩĩw'e nĩ ing'ei sya mtandaonĩ.